Kodewala Academy – Java Backend Tests

Complete Question Bank

Test 3 – Java Backend Test # 3 (45 questions)

Test 3 – Java Backend Test # 3 — 45 questions

Q1. Which statement is true about JDK?

A. Contains only JVM
B. Contains only JRE
C. Contains only compiler
D. Contains JRE, JVM, and development tools

Correct answer: D. Contains JRE, JVM, and development tools
Explanation: JDK = JRE (which includes JVM) + development tools such as javac.

Q2. Why is Java called platform independent?

A. Java code is interpreted directly by OS
B. Bytecode can run on any JVM
C. Java programs require no OS
D. JVM is platform independent

Correct answer: B. Bytecode can run on any JVM
Explanation: Source compiles to bytecode, which runs on any platform having a JVM (WORA).

Q3. What is the output?

```
int i = 1;
while(i <= 3){
    System.out.print(i++);
}
```

A. 0123
B. 123
C. 1234
D. Infinite loop

Correct answer: B. 123
Explanation: Prints 1, 2, 3 (post-increment), then i becomes 4 and the loop ends.

Q4. Which loop executes at least once?

A. for
B. while
C. do-while
D. enhanced for loop

Correct answer: C. do-while
Explanation: do-while checks its condition after the body.

Q5. Which operator short-circuits evaluation?

A. &
B. Both || and &&
C. ||
D. &&

Correct answer: B. Both || and &&
Explanation: && and || skip the right operand when the result is already known; & doesn't.

Q6. What is stored in args[0]?

```
// java Test Hello
```

A. Test
B. java
C. Hello
D. args

Correct answer: C. Hello
Explanation: args holds only the arguments after the class name, so args[0] is Hello.

Q7. Which member belongs to the class rather than an object?

A. Instance Variable
B. Static Variable
C. Local
D. None of the above

Correct answer: B. Static Variable
Explanation: Static members belong to the class.

Q8. Which method signature is valid?

A. void test(int a)
B. int test()
C. public static void main(String[] args)
D. All of the above

Correct answer: D. All of the above
Explanation: All three are syntactically valid method declarations.

Q9. Can two methods have the same name?

A. Never
B. Only in different classes
C. Yes, through method overloading
D. No

Correct answer: C. Yes, through method overloading
Explanation: Methods with the same name but different parameter lists are overloaded.

Q10. Which access modifier gives the widest accessibility?

A. protected
B. private
C. public
D. default

Correct answer: C. public
Explanation: public is accessible from everywhere.

Q11. Which access modifier restricts access within the same class only?

A. protected
B. private
C. public
D. default

Correct answer: B. private
Explanation: private is limited to the declaring class.

Q12. What is the output?

```
for(int i=1;i<=5;i++){
    if(i==3)
    {
        continue;
    }
    System.out.print(i);
}
```

A. 12345
B. 1245
C. 345
D. 12

Correct answer: B. 1245
Explanation: continue skips the print when i == 3, giving 1245.

Q13. What is the output?

```
for(int i=1;i<=5;i++){
    if(i==3)
    {
        break;
    }
    System.out.print(i);
}
```

A. 12345
B. 1245
C. 345
D. 12

Correct answer: D. 12
Explanation: break exits the loop at i == 3, so only 1 and 2 are printed.

Q14. Which statement immediately exits a method?

A. break
B. continue
C. exit
D. return

Correct answer: D. return
Explanation: return ends method execution and returns to the caller.

Q15. Which package is automatically imported?

A. java.util
B. java.io
C. java.lang
D. java.sql

Correct answer: C. java.lang
Explanation: java.lang (String, System, Object...) is imported implicitly.

Q16. Which statement about packages is correct?

A. Package names must start with capital letters
B. Packages help avoid naming conflicts
C. Package names can contain spaces
D. Packages are optional folders

Correct answer: B. Packages help avoid naming conflicts
Explanation: Packages namespace classes, avoiding name clashes and organising code.

Q17. What is the size of the array?

```
int[] arr = new int[5];
```

A. 4
B. 5
C. 6
D. Depends on JVM

Correct answer: B. 5
Explanation: new int[5] allocates 5 elements (indexes 0-4).

Q18. What is the default value of an int array element?

A. null
B. 1
C. 0
D. Undefined

Correct answer: C. 0
Explanation: Numeric arrays are zero-initialised.

Q19. What happens?

```
int[] arr = new int[3];
System.out.println(arr[3]);
```

A. 0
B. null
C. ArrayIndexOutOfBoundsException
D. Compilation Error

Correct answer: C. ArrayIndexOutOfBoundsException
Explanation: Valid indexes are 0-2. Index 3 throws an exception at runtime.

Q20. Which is NOT a JVM memory area?

A. Heap
B. stack
C. metaspace
D. Database Area

Correct answer: D. Database Area
Explanation: JVM areas include Heap, Stack, Metaspace/Method Area, PC registers, native stack. There is no 'Database Area'.

Q21. What is the output?

```
public class Test{
    static int count;
    public static void main(String[] args){
        System.out.println(count);
    }
}
```

A. null
B. 0
C. 1
D. Compilation Error

Correct answer: B. 0
Explanation: Static int fields default to 0.

Q22. Coupon Validation: what is the output?

```
boolean isPremium = false;
boolean couponValid = true;
if(isPremium || couponValid)
{
    System.out.println("Discount Applied");
}
```

A. No Output
B. Discount Applied
C. Compilation Error
D. Runtime Error

Correct answer: B. Discount Applied
Explanation: false || true is true, so the message prints.

Q23. Which class can access stock directly?

```
public class Inventory{
    private int stock = 10;
}
```

A. Any class
B. Same package
C. Subclass
D. Inventory class only

Correct answer: D. Inventory class only
Explanation: private limits access to the declaring class.

Q24. What is the output?

```
int[] ids = {101,102,103};
System.out.println(ids[3]);
```

A. 103
B. null
C. ArrayIndexOutOfBoundsException
D. 0

Correct answer: C. ArrayIndexOutOfBoundsException
Explanation: The array has 3 elements (indexes 0-2), so ids[3] throws an exception.

Q25. A customer can get a discount if they are a premium member OR they have a valid coupon. Why is validCoupon not evaluated?

```
boolean premiumMember = true;
boolean validCoupon = false;
if(premiumMember || validCoupon)
{
    System.out.println("Discount Applied");
}
```

A. JVM Optimization
B. Compiler Error
C. Short-Circuit Evaluation of ||
D. Both variables are boolean

Correct answer: C. Short-Circuit Evaluation of ||
Explanation: premiumMember is already true, so || short-circuits.

Q26. Which key should be used if you want to see the execution of return a + b; when the debugger is stopped at int result = add(10, 20);?

```
public static void main(String[] args)
{
    int result = add(10, 20);
    System.out.println(result);
}
public static int add(int a, int b)
{
    return a + b;
}
```

A. F8
B. F7
C. F6
D. F5

Correct answer: D. F5
Explanation: F5 (Step Into) enters the add() method.

Q27. You are debugging int result = add(10,20);. You don't want to enter the add() method. Which key should you use?

A. F8
B. F7
C. F6
D. F5

Correct answer: C. F6
Explanation: F6 (Step Over) executes the call without entering the method.

Q28. What does F8 do?

A. Enter method
B. Execute current line only
C. Run till next breakpoint or program completion
D. Exit JVM

Correct answer: C. Run till next breakpoint or program completion
Explanation: F8 = Resume.

Q29. Which variable is stored in Stack Memory during execution?

```
class Product {
    static int gst = 18;
    int price = 1000;
    public void calculate() {
        int discount = 100;
    }
}
```

A. gst
B. price
C. discount
D. Product object

Correct answer: C. discount
Explanation: discount is a local variable (stack). gst is static (method area/metaspace) and price lives in the object on the heap.

Q30. An inventory system stores 10,000 product IDs in an array. Which characteristic of arrays allows quick retrieval of any element using its index?

A. Elements are sorted automatically.
B. Elements are stored in contiguous memory locations.
C. Arrays use HashMap internally.
D. Arrays store data in Method Area.

Correct answer: B. Elements are stored in contiguous memory locations.
Explanation: Contiguous storage gives O(1) access: address = base + index × size.

Q31. If arrays require contiguous memory, what happens when we need a larger array?

A. JVM automatically increases the same memory block.
B. A new larger array is created and data is copied.
C. Existing array size changes.
D. Array moves to Stack.

Correct answer: B. A new larger array is created and data is copied.
Explanation: Array size is fixed, so you allocate a bigger array and copy elements over.

Q32. When should a for loop be preferred over a while loop?

A. When the number of iterations is unknown
B. Only for arrays
C. Only for strings
D. When the number of iterations is known in advance

Correct answer: D. When the number of iterations is known in advance
Explanation: for bundles init, condition and update, which suits a known iteration count.

Q33. Which loop is generally preferred for reading user input until the user enters a specific value?

A. for loop
B. while loop
C. do-while
D. none

Correct answer: B. while loop
Explanation: The iteration count is unknown, so a condition-driven while loop fits.

Q34. What happens?

```
for( ; ; )
{
    System.out.println("Java");
}
```

A. Prints once
B. Compilation Error
C. Infinite Loop
D. Runtime Exception

Correct answer: C. Infinite Loop
Explanation: With no condition, for(;;) runs forever.

Q35. Which statement is TRUE?

A. for loop cannot become an infinite loop.
B. while loop cannot become an infinite loop.
C. Both for and while can become infinite loops.
D. Neither can become an infinite loop.

Correct answer: C. Both for and while can become infinite loops.
Explanation: Any loop whose condition never becomes false (or is omitted) runs forever.

Q36. An ATM application keeps asking for a PIN until the correct PIN is entered. Which loop is more suitable?

A. for
B. while
C. do-while
D. switch

Correct answer: B. while
Explanation: The number of attempts is unknown, so a condition-based loop fits.

Q37. An inventory system needs to scan exactly 500 products. Which loop should be preferred?

A. while loop because it is faster
B. for loop because the number of iterations is known
C. switch case
D. do-while

Correct answer: B. for loop because the number of iterations is known
Explanation: A known iteration count suits for; speed is not the reason.

Q38. An inventory system needs to scan exactly 500 products. Which loop should be preferred? (duplicate)

A. while loop because it is faster
B. for loop because the number of iterations is known
C. switch case
D. do-while

Correct answer: B. for loop because the number of iterations is known
Explanation: Same as Q37: use for when the iteration count is known.

Note: This question appears twice (Q37 and Q38) in the test; the stem is identical.

Q39. What is the output?

```
class A {
    A() { System.out.println("A"); }
    A(int x) { this(); System.out.println("A-int"); }
}
class B extends A {
    B() { this(10); System.out.println("B"); }
    B(int x) { super(x); System.out.println("B-int"); }
}
public class Test {
    public static void main(String[] args) { new B(); }
}
```

A. A → A-int → B-int → B
B. A-int → A → B → B-int
C. A → B-int → A-int → B
D. Compilation error

Correct answer: A. A → A-int → B-int → B
Explanation: B() → this(10) → B(int) → super(x) → A(int) → this() → A() prints A; then A-int, B-int, B.

Q40. What is the output?

```
int total = 0;
for (int i = 1; i <= 5; i++)
{
    if (i % 2 == 0)
    {
        continue;
    }
    total += i;
    if (total > 4)
    {
        break;
    }
}
System.out.println(total);
```

A. 1
B. 4
C. 5
D. 9

Correct answer: D. 9
Explanation: i=1 → total 1; i=2 skipped; i=3 → total 4 (not > 4); i=4 skipped; i=5 → total 9 > 4 → break. Prints 9.

Q41. Output?

```
class Parent {
    Parent() { System.out.println("Parent"); }
}
class Child extends Parent {
    Child() { this(10); System.out.println("Child"); }
    Child(int x) { System.out.println("Child-int"); }
}
public class Test {
    public static void main(String[] args) { new Child(); }
}
```

A. Child → Child-int → Parent
B. Parent → Child-int → Child
C. Child-int → Parent → Child
D. Compilation error

Correct answer: B. Parent → Child-int → Child
Explanation: Child() delegates to Child(int), which implicitly calls super(). Order: Parent, Child-int, Child.

Q42. Which statement about this() and super() is correct?

A. Both can be used together in the same constructor
B. this() must always be the last statement
C. this() or super() must be the first statement of a constructor
D. super() can be called from any method

Correct answer: C. this() or super() must be the first statement of a constructor
Explanation: A constructor call must be the first statement, and only one of the two can be used.

Q43. What happens?

```
class A {
    A() { this(10); }
    A(int x) { this(); }
}
```

A. Executes successfully
B. Runtime exception
C. Compile-time error due to recursive constructor invocation
D. StackOverflowError at runtime

Correct answer: C. Compile-time error due to recursive constructor invocation
Explanation: Constructors calling each other in a cycle are detected by the compiler (recursive constructor invocation).

Q44. What happens?

```
class A {
    A(int x) { System.out.println("A"); }
}
class B extends A {
    B() { System.out.println("B"); }
}
```

A. A → B
B. B only
C. Compilation error
D. Runtime exception

Correct answer: C. Compilation error
Explanation: B() implicitly calls super(), but A has no no-arg constructor (only A(int)), so compilation fails.

Q45. What will be printed, and what will be the final values of orderId and status?

```
class Order {
    int orderId;
    String status;
    Order() { this(0); System.out.println("Default Order"); }
    Order(int orderId) { this(orderId, "NEW"); System.out.println("Order with ID"); }
    Order(int orderId, String status) {
        this.orderId = orderId;
        this.status = status;
        System.out.println("Order Created");
    }
}
public class Test {
    public static void main(String[] args) { Order order = new Order(); }
}
```

A. Order Created / Order with ID / Default Order / orderId = 0 / status = "NEW"
B. Default Order / Order with ID / Order Created / orderId = 0 / status = "NEW"
C. Order Created / Default Order / Order with ID / orderId = 0 / status = null
D. Compilation error because constructors cannot call another constructor

Correct answer: A. Order Created / Order with ID / Default Order / orderId = 0 / status = "NEW"

Explanation: new Order() → this(0) → this(0,"NEW"). The innermost runs first (Order Created), then Order with ID, then Default Order. Final values: orderId = 0, status = "NEW".